Korea Technology and Information Promotion Agency for SMEs

(max 10 pages)

Technology Proposal for 2026 TIPA* International Joint R&D Program

(TIPA: Korea Technology and Information Promotion Agency for SMEs)

The objective of this Technology Proposal is to introduce the institute's proprietary technologies and present potential collaborative areas for Korean SMEs, facilitating joint research and development projects under the 2026 TIPA International Joint R&D Program. This Technology Proposal will be publicly disclosed in conjunction with the official announcement of the 2026 TIPA International Joint R&D Program.

1. Technology Proposer Information

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2. Technology Overview

Title of Proposed Technology	AI based development of ionizable lipids and lipid nanoparticle formulation for CNS delivery of Base Editors			
Korea's 12 National Strategic Technology Fields	Semiconductor/Display (), Secondary Battery ( ), Advanced Mobility ( ), Next-gen Nuclear Power ( ), Advanced Bio (√), Aerospace/Marine ( ), Hydrogen ( ), Cybersecurity ( ), AI (√), Next-gen Communication ( ), Advanced Robotics & Manufacturing ( ), Quantum ( )			
Industrial Technology Classification [1)]	**Technology Field 1** (Large Category)		**Technology Field 2** (Large Category)	
	Bio・Medical		Bio・Medical	
	Medium Category	Small Category	Medium Category	Small Category
	Medical Bio	Drug Delivery Systems	Medical Bio	Drug Delivery Systems
Concept of Proposed Technology	The proposed technology will establish a next-generation lipid nanoparticle (LNP) platform for efficient central nervous system (CNS) delivery of gene editing therapeutics. By integrating AI-guided ionizable lipid design with advanced formulation and manufacturing strategies, the project targets a central limitation in CNS gene therapy: achieving potent, cell-selective delivery to neuronal and glial populations. The platform will be optimized for intrathecal administration and engineered to co-deliver base editor mRNA together with guide RNAs, enabling robust in situ editing in the CNS while minimizing off-target biodistribution. The resulting delivery system is intended to be modular and broadly transferable across neurodegenerative and monogenic CNS disorders. As an initial proof of concept, Alzheimer's disease will be used to validate therapeutic potential via base editor-mediated modulation of amyloid precursor protein (APP) processing pathways.			

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Rationale for Joint R&D Project	The Rationale for Joint R&D Project centers on the critical need to bridge the gap between advanced gene-editing technologies and effective delivery systems for the central nervous system (CNS). While significant progress has been achieved in the development of base editors, their clinical application for neurological disorders is fundamentally limited by the blood-brain barrier and the structural complexity of the brain. Current non-viral delivery methods often fail to achieve the deep parenchymal penetration required for therapeutic impact, and viral vectors pose long-term safety and scalability concerns. This project addresses these bottlenecks by integrating the University of Toronto's AI-driven lipid discovery platform with a Korean SME's specialized capabilities in mRNA design and translational manufacturing. The University of Toronto team has established a robust foundation for this partnership through the discovery of lead ionizable lipids, such as P3B, which demonstrate a 30-fold increase in brain-specific delivery efficiency compared to industry benchmarks like DLin-MC3-DMA. This discovery provides a high-performance, biodegradable starting point for therapeutic development, yet the transition to clinical-grade applications requires the specialized expertise of an industrial partner. A potential Korean SME partner is expected to contribute essential strengths in the precision design of base editor mRNA, optimization of large-scale LNP characterization, and the establishment of disease-relevant models for therapeutic validation. This joint R&D framework creates a synergistic pipeline where academic innovation in molecular bioengineering and AI-guided synthesis is directly supported by industrial manufacturing and regulatory pathways. By combining these complementary skill sets, the consortium can accelerate the development of a CNS-targeting platform capable of precise single-nucleotide correction for neurodegenerative diseases. This collaboration not only advances Korea's technological standing in the advanced bio field but also establishes a scalable model for translating high-impact academic research into life-changing RNA therapeutics
Key Words	1. Ionizable lipid nanoparticles 2. CNS drug delivery 3. Base editor 4. AI-guided lipid design

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3. Research and Development Objectives & Plan

Technology Readiness Level (TRL)[2]	Start of the Project	Level (2)	End of the Project	Level (6)
R&D Objectives	The primary objective of this project is to establish an optimized lipid nanoparticle platform capable of delivering base editors to neuronal cells through intrathecal administration. Specific objectives include: • Identification of novel ionizable lipids using AI-guided design and in vivo screening • Optimization of LNP formulation for CNS distribution and safety • Validation of base editing efficiency in disease-relevant CNS models • Evaluation of translational feasibility for future clinical development			
R&D Target Specifications	• Efficient neuronal transfection following intrathecal administration • Demonstrated base editing activity in CNS tissues • Acceptable safety and tolerability profiles in animal models • Reproducible formulation suitable for scale-up studies			
R&D Plan	Year 1 • AI-based ionizable lipid candidate selection • Initial LNP formulation development • In vivo CNS distribution and safety evaluation Year 2 • Optimization of lipid structures and formulation parameters • Neuron- and glial-cell targeting assessment • Base editor delivery and editing efficiency validation Year 3 • Therapeutic evaluation in Alzheimer's disease or related CNS models • Scale-up formulation feasibility study • Preparation for translational and commercialization pathways			

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4. R&R (Roles and Responsibilities)

Expected R&R of Research Institute	• AI-guided design and synthesis of novel ionizable lipids • Development of lipid libraries and structure–function analysis • In vivo CNS delivery evaluation and editing analysis • Scientific guidance on CNS gene delivery strategies
Expected R&R of Korean SME Partner	• Design and production of base editor mRNA and guide RNAs • LNP formulation optimization and characterization • Disease model establishment and therapeutic evaluation • Scale-up formulation and translational feasibility studies

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5. Expected Outcomes/Impact

Expected Outcomes and Impact	The successful execution of this joint R&D project will establish a robust, non-viral, CNS-targeting lipid nanoparticle platform capable of delivering large-cargo gene editing therapeutics with unprecedented efficiency. By validating this technology through the precise modulation of APP processing, the project will provide a translatable framework for treating Alzheimer's disease and other single-nucleotide variant-driven neurodegenerative disorders. This platform is expected to achieve significant on-target adenine base editing in vivo, building upon the established benchmark of 14.8% correction efficiency demonstrated by the P3B lipid system. The outcomes of this collaboration will facilitate the advancement of gene therapy research and foster deep, sustained scientific integration between Korean research-driven SMEs and leading global academic institutions like the University of Toronto. By transitioning the technology from a laboratory proof-of-concept (TRL 2) to a pilot-scale prototype (TRL 6), the project will directly strengthen Korea's technological leadership in the advanced biology and genomic medicine sectors. Ultimately, these efforts will provide a validated, biodegradable, and safe foundation for future translational clinical trials, bridging the gap between innovative lipid chemistry and life-saving RNA therapeutics in the CNS field.

6. References

(1) Dong, Songtao, et al. "Biodegradable lipid nanoparticles for genome editing in the brain via intrathecal administration." Materials Today (2025).

(2) Xu, Yue, et al. "AGILE platform: a deep learning powered approach to accelerate LNP development for mRNA delivery." Nature communications 15.1 (2024): 6305.

(3) Guyon, Antoine, et al. "Base editing strategy for insertion of the A673T mutation in the APP gene to prevent the development of AD in vitro." Molecular Therapy Nucleic Acids 24 (2021): 253-263.

(4) Chang, Jennifer L., et al. "Targeting amyloid-β precursor protein, APP, splicing with antisense oligonucleotides reduces toxic amyloid-β production." Molecular Therapy 26.6 (2018): 1539-1551.

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☐ Terms:

1) Industrial Technology Classification

· Classification system according to Industrial Technology Classification Table (**Appendix 1**, based on Article 16 of the Common Operational Guidelines for the Industrial Technology Innovation Project in Korea).

2) Technology Readiness Level (TRL)

Description		
·A consistent and objective indicator of the technological maturity of key element technologies		
√	**Basic Research Stage (1-2)**	1st Stage: Basic theories and experiments 2nd Stage: Proof of Concept, such as practical ideas or patents
	Experimental Stage (3-4)	3rd Stage: Verification of basic performance at a laboratory scale 4th Stage: Evaluation of key performance of materials, components, and systems at a laboratory scale
	Prototype Stage (5-6)	5th Stage: Production and performance evaluation of prototypes of finalized materials, components, and systems 6th Stage: Production and performance evaluation of prototypes at a pilot scale
	Productization Stage (7-8)	7th Stage: Reliability testing and evaluation by potential customers 8th Stage: Prototype certification and standardization
	Commercialization Stage (9)	9th Stage: Commercialization

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